CS 6515: Graduate Algorithms

Lecture 10: LP Duality and Max-Flow/Min-Cut

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Announcements. HW3 is out; due Oct 13th.

Today's plan. We begin with the optimality-certificate discussion left over from Lecture 9, then derive LP duality and apply it to max-flow/min-cut.

1 How do we prove a proposed solution is optimal?

Recall the smoothie LP from Lecture 9:

$$\begin{aligned} \text{maximize} \quad & 5x_1 + 4x_2 + 6x_3 \\ \text{subject to} \quad & x_1 + x_3 \le 3 \qquad \text{(oranges)}, \\ & 2x_1 + x_2 + x_3 \le 10 \qquad \text{(bananas)}, \\ & 2x_2 + x_3 \le 6 \qquad \text{(strawberries)}, \\ & x_1, x_2, x_3 \ge 0. \end{aligned}$$

The solution $x = (3, 3, 0)$ uses 3 oranges, 9 bananas, and 6 strawberries, so it is feasible and has value $5 \cdot 3 + 4 \cdot 3 = 27$. This gives a *lower bound* on the maximum. To prove optimality, we need an upper bound of 27 for *every* feasible solution.

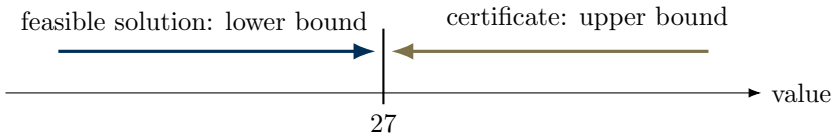
Build an upper bound from the constraints. Take five times the orange constraint and two times the strawberry constraint:

$$5(x_1 + x_3) + 2(2x_2 + x_3) \le 5 \cdot 3 + 2 \cdot 6 = 27.$$

The left side is $5x_1 + 4x_2 + 7x_3$. Since $x_3 \ge 0$,

$$\boxed{5x_1 + 4x_2 + 6x_3 \le 5x_1 + 4x_2 + 7x_3 \le 27.}$$

Equivalently, include one copy of $-x_3 \le 0$ to make the left side exactly the objective. The matching lower and upper bounds prove that $x = (3, 3, 0)$ is optimal.



From one certificate to a general method. The weights $(5, 0, 2)$ give a short *certificate* of optimality. How could we systematically find the best such upper bound? We will treat the nonnegative constraint weights as variables: the search for a certificate is itself an LP, called the **dual**.

2 Deriving the dual: optimize over upper bounds

The smoothie argument used only two facts: nonnegative weighted sums preserve inequalities, and the weighted left side dominates the objective coefficient by coefficient. Let us generalize it to the canonical primal LP

$$(P) \qquad \max\ c^\top x \quad \text{subject to} \quad Ax \le b, \quad x \ge 0.$$

Put a nonnegative weight y_j on row j and add the inequalities:

$$\sum_{j=1}^{m} y_j \left(\sum_{i=1}^{n} A_{ji} x_i \right) \le \sum_{j=1}^{m} y_j b_j.$$

If $\sum_j A_{ji} y_j \ge c_i$ for every variable i, then nonnegativity of x_i gives

$$\boxed{c^\top x \ \le\ x^\top A^\top y \ =\ y^\top A x \ \le\ b^\top y.}$$

Thus every such y certifies an upper bound $b^\top y$. To find the best certificate, solve

$$(D) \qquad \min\ b^\top y \quad \text{subject to} \quad A^\top y \ge c, \quad y \ge 0.$$

Primal maximization LP	Dual minimization LP
n variables x_i	n constraints $(A^\top y)_i \ge c_i$
m constraints $(Ax)_j \le b_j$	m variables y_j
Objective coefficients c_i	Right-hand sides c_i
Right-hand sides b_j	Objective coefficients b_j

Why the signs matter. We need $y \ge 0$ when multiplying the primal inequalities, and $x \ge 0$ when comparing $c^\top x$ with $(A^\top y)^\top x$. Neither sign condition is cosmetic.

3 Weak duality and a worked certificate

Theorem (weak LP duality). For every primal-feasible x and dual-feasible y,

$$c^\top x \le b^\top y.$$

Consequently, if $c^\top x = b^\top y$, then both solutions are optimal.

Proof. The inequality is the chain in the preceding section. Equality leaves no room for either LP to improve.

3.1 Revisiting the smoothie certificate

The three resource weights from our opening example now become variables y_O, y_B, y_S. Applying the general construction gives

$$\begin{array}{ll} \text{minimize} & 3y_O + 10y_B + 6y_S \\ \text{subject to} & y_O + 2y_B \ge 5, \\ & y_B + 2y_S \ge 4, \\ & y_O + y_B + y_S \ge 6, \\ & y_O, y_B, y_S \ge 0. \end{array}$$

The first constraint corresponds to smoothie 1, the second to smoothie 2, and the third to smoothie 3.

Example. Our opening weights $y = (5, 0, 2)$ satisfy all three dual constraints: $5 \ge 5$, $4 \ge 4$, and $7 \ge 6$. Their dual value is $3 \cdot 5 + 10 \cdot 0 + 6 \cdot 2 = 27$, matching the primal value at $x = (3, 3, 0)$. The earlier certificate is exactly an application of weak duality: both solutions are optimal.

Economic interpretation. Imagine purchasing all available ingredients at prices y_O, y_B, y_S. Every dual constraint says that the ingredient cost of one liter of a smoothie is at least its revenue. Therefore no production plan can earn more than the total priced inventory. The dual minimizes this valid upper bound.

At the displayed optimum, one banana is unused and the banana price is zero. Smoothie 3 has priced ingredients worth 7, exceeding revenue 6, and is not produced. These observations will become *complementary slackness* in Lecture 11.

Exercise. Check that $(y_O, y_B, y_S) = (3, 1, 2)$ is dual-feasible. What upper bound does it give? Why is a feasible certificate not necessarily the best certificate?

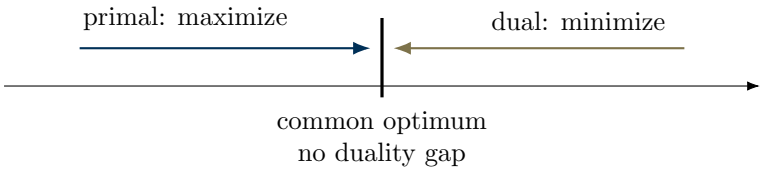
4 Strong duality and other LP forms

Theorem (strong LP duality). If the primal LP is feasible with a finite optimal value, then the dual also has an optimum, and

$$\boxed{\max\{c^\top x : Ax \le b,\ x \ge 0\} = \min\{b^\top y : A^\top y \ge c,\ y \ge 0\}.}$$

The symmetric statement holds starting from a finite dual optimum.

Weak duality says every certificate is an upper bound. Strong duality says an *exact* certificate exists. We will prove this in Lecture 11.



Infeasibility is not the same as unboundedness.

- If the primal is unbounded above, the dual is infeasible.
- If the dual is unbounded below, the primal is infeasible.
- The two LPs can *both* be infeasible. Thus the converse of either preceding statement is false.

For example, the primal

$$\max(2x_1 - x_2): \quad x_1 - x_2 \le 1, \quad -x_1 + x_2 \le -2, \quad x \ge 0$$

is infeasible. Its dual requires $y_1 - y_2 \ge 2$ and $-y_1 + y_2 \ge -1$, which is also impossible.

4.1 A compact rulebook

For a *maximization* primal and its minimization dual:

Primal feature	Corresponding dual feature
Constraint $a_j^\top x \le b_j$	$y_j \ge 0$
Constraint $a_j^\top x \ge b_j$	$y_j \le 0$
Constraint $a_j^\top x = b_j$	y_j unrestricted
Variable $x_i \ge 0$	$(A^\top y)_i \ge c_i$
Variable $x_i \le 0$	$(A^\top y)_i \le c_i$
Variable x_i unrestricted	$(A^\top y)_i = c_i$

The objective is still $b^\top y$. Derive these rules by preserving the weak-duality inequality, or first convert to canonical form. The dual of the dual is the primal.

5 Application: max-flow/min-cut from LP duality

Lecture 8 used residual reachability to produce a cut with the same value as a maximum flow. We now obtain the same equality by a different route.

5.1 A path-based flow LP

Let $\mathcal{P}$ be the finite set of all simple directed s-t paths. Give path p a variable x_p, the amount sent along it:

$$\begin{aligned} (P_{\text{flow}}) \quad \text{maximize} \quad & \sum_{p \in \mathcal{P}} x_p \\ \text{subject to} \quad & \sum_{p \ni e} x_p \le c_e \quad (e \in E), \\ & x_p \ge 0 \quad (p \in \mathcal{P}). \end{aligned}$$

Adding the path flows gives an ordinary feasible flow of the same value. Conversely, any feasible flow of nonnegative value can be decomposed into s-t path flows and directed cycle flows; discard the cycles without changing its value. One way to see the decomposition is to add an edge $t \to s$ carrying the flow value and decompose the resulting circulation into cycles.

Size warning. There can be exponentially many simple paths. This formulation is for understanding duality, not an instruction to enumerate all paths. The edge-based LP in Lecture 9 has polynomial size.

5.2 The dual assigns edge lengths

There is a dual variable y_e for each capacity constraint:

$$\begin{aligned} (D_{\text{length}}) \quad \text{minimize} \quad & \sum_{e \in E} c_e y_e \\ \text{subject to} \quad & \sum_{e \in p} y_e \ge 1 \quad (p \in \mathcal{P}), \\ & y_e \ge 0 \quad (e \in E). \end{aligned}$$

Interpret y_e as a length: every s-t path must have length at least 1. Strong duality equates this fractional length optimum with maximum flow.

Cuts give feasible dual solutions. For an s-t cut S, set $y_e = 1$ on edges leaving S and 0 elsewhere. Every s-t path leaves S at least once, and the dual value is exactly the capacity $c(\delta^+(S))$, where $\delta^+(S) = \{(u, v) \in E : u \in S,\ v \notin S\}$.

The missing step is important: *why can a fractional length solution be replaced by a cut without increasing its cost?*

6 From fractional lengths to an actual cut

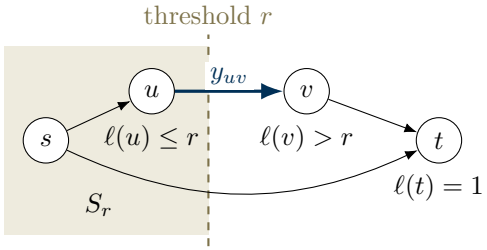
Take any feasible dual lengths $y \geq 0$. Let $d(v)$ be shortest-path distance from s in these lengths, with $d(v) = +\infty$ if unreachable. Define clipped distances

$$\ell(v) = \min\{d(v), 1\}, \qquad \ell(s) = 0, \qquad \ell(t) = 1.$$

The last equality holds because every s-t path has length at least 1. For each threshold $r \in (0, 1)$, define

$$S_r = \{v : \ell(v) \leq r\}.$$

Every S_r is an s-t cut.



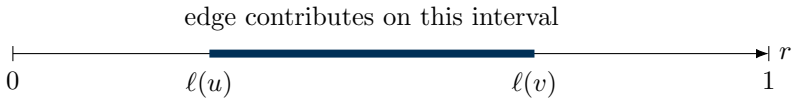
Only directed edges leaving S_r contribute to its capacity.

6.1 For how many thresholds is one edge cut?

An edge $e = (u, v)$ leaves S_r exactly when $\ell(u) \leq r < \ell(v)$. The length of this interval is

$$\max\{0, \ell(v) - \ell(u)\} \leq y_{uv}.$$

Indeed, when $d(u) < 1$, the shortest-path inequality gives $d(v) \leq d(u) + y_{uv}$, and clipping cannot increase the positive difference. When $d(u) \geq 1$ or u is unreachable, $\ell(u) = 1$ and the positive difference is zero.



6.2 Average the cut capacities

Sweep r from 0 to 1. The average cut capacity is at most

$$\begin{aligned}\int_0^1 c(\delta^+(S_r))\,dr &= \sum_{(u,v)\in E} c_{uv}\max\{0, \ell(v)-\ell(u)\} \\ &\le \sum_{e\in E} c_e y_e.\end{aligned}$$

This integral is just the area of a step function: the cut changes only when r passes one of the finitely many vertex distances. Equivalently, take a weighted average of the finitely many cut values, with weights equal to the lengths of the threshold intervals.

At least one cut is no more expensive than the average. Therefore

$$\boxed{\text{some } S_r \text{ satisfies } c(\delta^+(S_r)) \le \sum_e c_e y_e.}$$

The cut can be found deterministically by checking the distinct threshold cuts after computing shortest-path distances.

6.3 Complete the proof

Apply the rounding argument to an optimal dual solution:

$$\begin{aligned}\text{maximum-flow value} &\le \text{minimum-cut capacity} && \text{(the cut bound from Lecture 8)},\\ \text{minimum-cut capacity} &\le \text{optimal length-dual value} && \text{(threshold rounding)},\\ \text{optimal length-dual value} &= \text{maximum-flow value} && \text{(strong LP duality)}.\end{aligned}$$

All inequalities must be equalities.

What is integral here? The length dual has an optimal 0/1 solution corresponding to a cut, even for nonnegative real capacities. This is different from the integrality of the *flow*, which in Lecture 8 required integral capacities.

Preview of approximation algorithms. Here rounding loses nothing. For other discrete problems, an LP may have a better fractional value than any integral solution. We will aim to round while losing only a bounded multiplicative factor.

Exercise. Given nonnegative lengths y_e, explain how one Dijkstra run can either verify every path constraint in D_{length}, or exhibit a violated one. Why do we not need to list all paths?